SETU SERVICES APP

Software Engineering Practice

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1. INTRODUCTION

This project is about creating an app called SETU Services App. The app helps students at university to ask for help and track their requests.

The app lets students ask for support with things like IT issues, facilities, library services. It also helps the university staff to manage and answer these requests quickly.

The main aim of the app is to make it easier for students and university services to talk to each other. It does this by providing a simple and easy to use platform where students can ask for help and get it.

The SETU Services App will help to improve communication, between students and university services. Students will be able to get the help they need quickly and easily. University staff will also be able to manage requests.

2. SYSTEM OVERVIEW / FEATURES

The SETU Services App is an application that helps students to access and manage university services efficiently. The system provides a centralized platform where students can submit service requests, track their progress, and book appointments with university departments.

The application supports two types of users: Students and Administrators. Students can interact with the system to request support, while Administrators manage and respond to those requests.

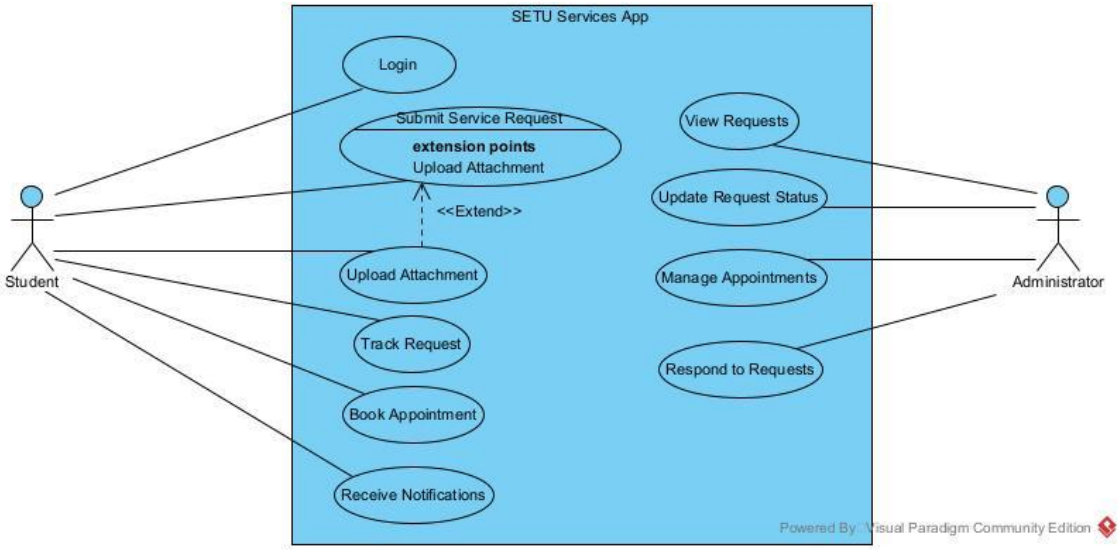
The main features of the system include:

- User authentication (login functionality)
- Submission of service requests
- Optional attachment upload when submitting requests
- Tracking the status of submitted requests
- Booking and managing appointments
- Receiving notifications about request updates
- Administrative management of requests and appointments

3. USE CASE DIAGRAM

The Use Case Diagram illustrates the interactions between the actors (Student and Administrator) and the system. Students can log in, submit service requests, upload attachments, track requests, and book appointments. Administrators can view requests, update request status, manage appointments, and respond to requests.

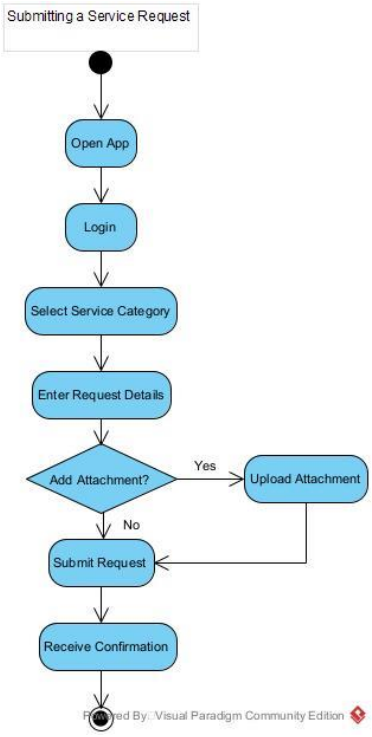
An <<extend>> relationship is used between “Submit Service Request” and “Upload Attachment” to indicate that uploading an attachment is optional.



4. ACTIVITY DIAGRAM

The Activity Diagram represents the workflow for submitting a service request. The process begins when the user opens the application and logs in. The user selects a service category and enters request details. A decision point allows the user to optionally upload an attachment. Finally, the request is submitted and a confirmation is received.

This diagram helps visualize the step-by-step flow of the system.

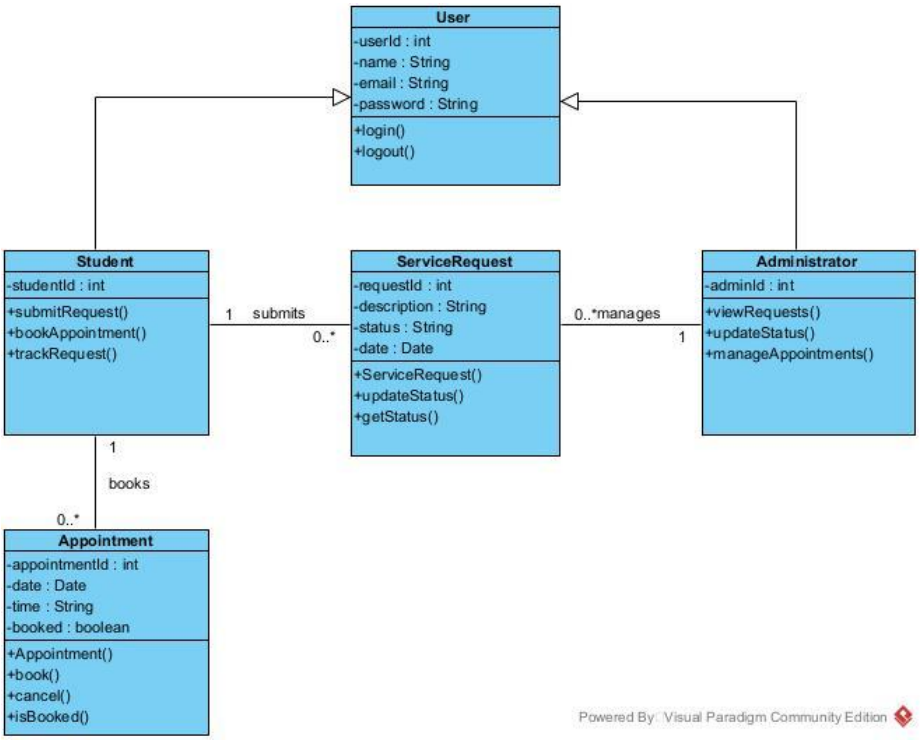


5. CLASS DIAGRAM

The Class Diagram models the structure of the system, including the main classes: User, Student, Administrator, ServiceRequest, and Appointment.

Student and Administrator inherit from the User class. A Student can submit multiple ServiceRequests and book multiple Appointments. An Administrator manages multiple ServiceRequests.

Each class contains attributes and methods that represent the system's behavior, such as updating request status and booking appointments.

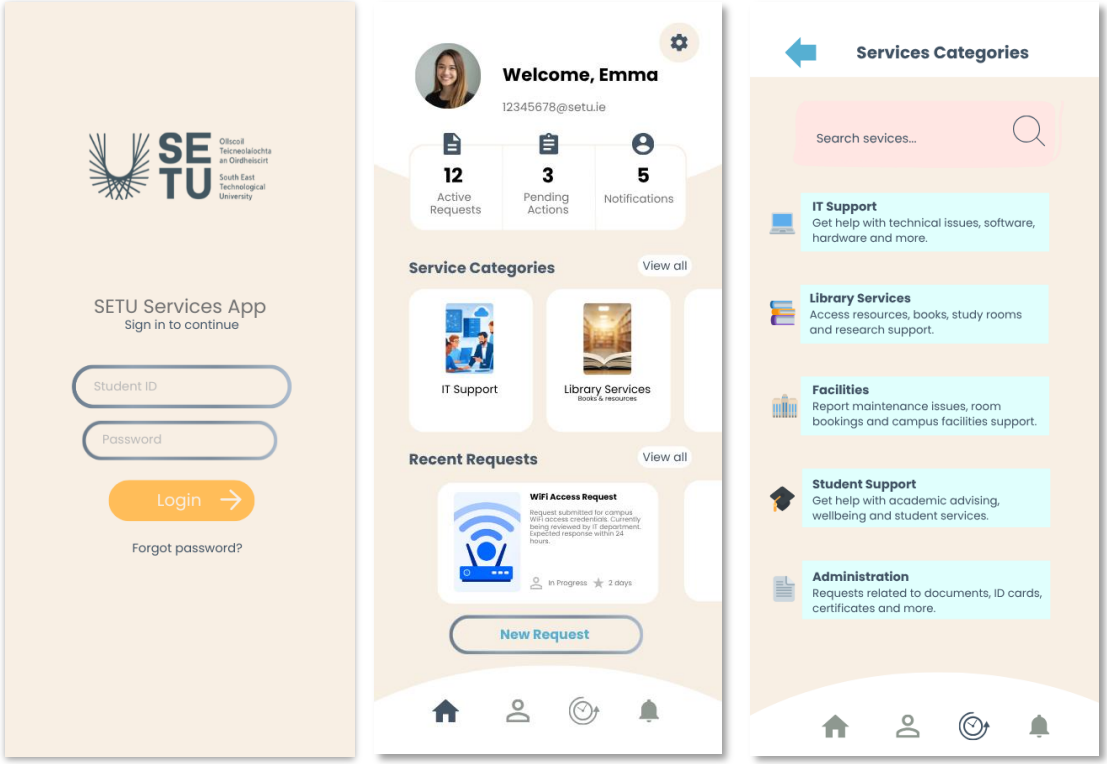


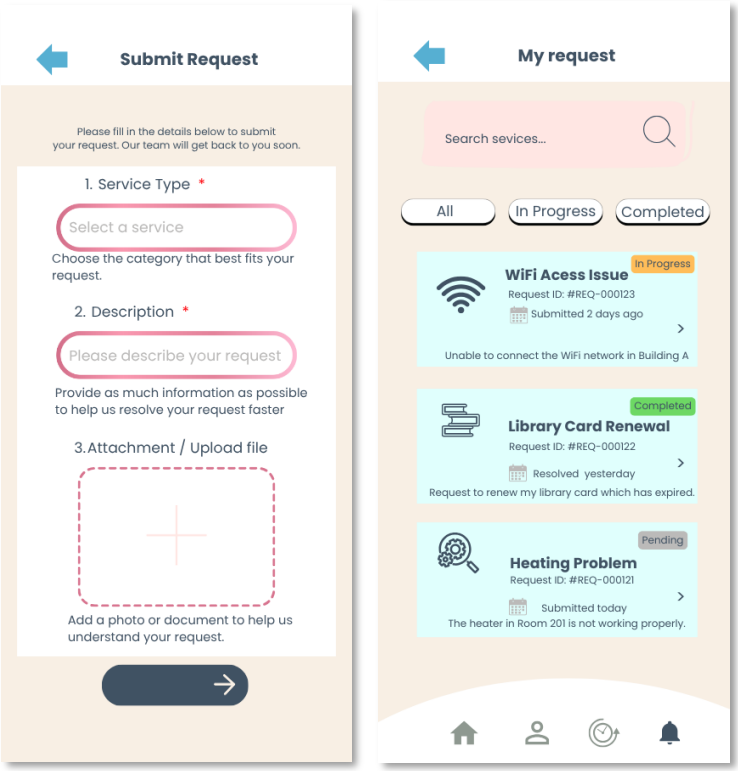
6. PROTOTYPE (UI DESIGN)

The prototype demonstrates the user interface of the SETU Services App. It includes key screens such as the login page, dashboard, service categories, request submission form, and request tracking page.

The design focuses on usability and simplicity, allowing students to easily navigate the application and perform tasks such as submitting requests and tracking their status. The administrator functionality is reflected through request management features.

The prototype helps visualize how users will interact with the system before implementation.





7. JUNIT TESTING

JUnit tests were implemented to verify the functionality of key components of the system.

Two main classes were tested:

1. ServiceRequest:
    - Verified that the initial status is "Pending"
    - Verified that the status updates correctly

ServiceRequest.java

```
package org.setu.services;

public class ServiceRequest {
    private String status;

    public ServiceRequest() {
        this.status = "Pending";
    }

    public void updateStatus(String newStatus) {
        this.status = newStatus;
    }

    public String getStatus() {
        return status;
    }
}
```

ServiceRequestTest.java

```
package org.setu.services;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class ServiceRequestTest {

    @Test
    void testInitialStatusIsPending() {
        ServiceRequest request = new ServiceRequest();

        assertEquals("Pending", request.getStatus());
    }

    @Test
```

```
    void testUpdateStatus() {
        ServiceRequest request = new ServiceRequest();
        request.updateStatus("Completed");

        assertEquals("Completed", request.getStatus());
    }

}
```

2. Appointment:
    - Verified that appointments are not booked initially
    - Verified booking functionality
    - Verified cancellation functionality

Appointment.java

```
package org.setu.services;

public class Appointment {
    private boolean booked;

    public Appointment() {
        this.booked = false;
    }

    public void book() {
        booked = true;
    }

    public void cancel() {
        booked = false;
    }

    public boolean isBooked() {
        return booked;
    }
}
```

AppointmentTest.java

```
package org.setu.services;
```

```java
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class AppointmentTest {

    @Test
    void testInitiallyNotBooked() {
        Appointment appt = new Appointment();

        assertFalse(appt.isBooked());
    }

    @Test
    void testBooking() {
        Appointment appt = new Appointment();
        appt.book();

        assertTrue(appt.isBooked());
    }

    @Test
    void testCancel() {
        Appointment appt = new Appointment();
        appt.book();
        appt.cancel();

        assertFalse(appt.isBooked());
    }
}
```

8. MARÍA DEL MAR REFLECTION

1. My Contribution

In this project, I contributed to the design and development of the SETU Services App. My main responsibilities included creating UML diagrams such as the Use Case Diagram, Activity Diagram, and Class Diagram. I also implemented the JUnit tests for key classes like ServiceRequest and Appointment.

Additionally, I worked on organizing the project structure, managing version control using GitHub, and ensuring that all components of the project were properly integrated.

2. What I Learned

Through this project, I developed a better understanding of UML modelling, including how to represent system behavior and structure using diagrams. I also learned how to write and execute JUnit tests to validate the functionality of Java classes.

Furthermore, I gained experience using Git and GitHub for version control, including committing and pushing changes. This helped me understand how collaborative development works in practice.

3. Challenges Faced

One of the main challenges I faced was understanding the correct use of UML relationships, particularly the difference between <<include>> and <<extend>> in Use Case Diagrams. It was initially difficult to decide when each should be used.

Another challenge was setting up the Java project and resolving issues related to JUnit and project configuration in IntelliJ. There were also some difficulties in connecting the local project to the GitHub repository.

4. How I solved them

To overcome these challenges, I reviewed UML concepts and applied them to my diagrams, making corrections where necessary. I also used online resources to resolve configuration issues in IntelliJ.

For GitHub integration, I followed step-by-step processes to correctly link the project and successfully push my work to the repository. These solutions helped me improve both my technical and problem-solving skills.

5. What I Would Improve

If I were to do this project again, I would spend more time planning the system design before starting implementation. This would help avoid making changes later in the process.

9. LAURA REFLECTION

1. My Contribution

In this project, my contributions were primarily towards UI design and prototyping for SETU Services through Canva, keeping usability and accessibility in mind during the entire process.
Moreover, I have designed user personas, identified key features, along with creating scenarios and user stories for proper requirement analysis.

2. **What I Learned**

This project helped me gain an appreciation of the concept of user-centric design and the ability to turn user requirements into system requirements. It helped me learn about creating user personas and user stories, which are used in designing applications.
In addition, I gained an appreciation for prototyping through the use of Canva, and I have seen the relationship between front-end design and system requirements. Testing was also emphasized through the use of JUnit.

3. **Challenges Faced**

One of the major issues I faced included coming up with user personas and user stories in such a way that they accurately reflect the needs of the actual user while still adhering to system requirements.
There were also difficulties when designing the interface. For instance, there was a need for me to make sure that it was not overly complicated but simple enough and at the same time still consistent with the prototype and system design.

4. **How I solved them**

In order to solve the above issues, I examined how other systems with similar functions have been designed, and used user-centric approaches that help come up with high-quality personas and user stories.
Also, I iterated through the design process of creating the prototype in order to come up with an improved design. In addition, collaboration with my teammate was key to ensuring consistency with the requirements.

5. **What I Would Improve**

If I were to do this assignment another time, one thing I would improve is the user story. This includes providing a lot more detail in order to assist with system design.
Another change will involve making the prototype more realistic and interactive in nature. Moreover, I would spend some time conducting user tests.